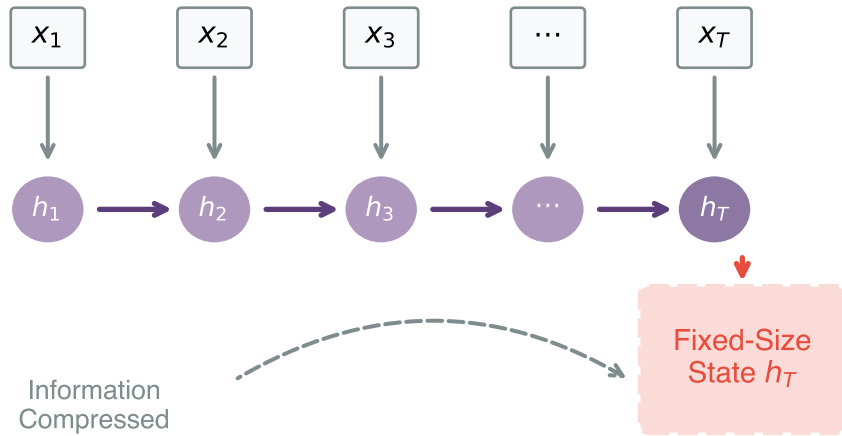
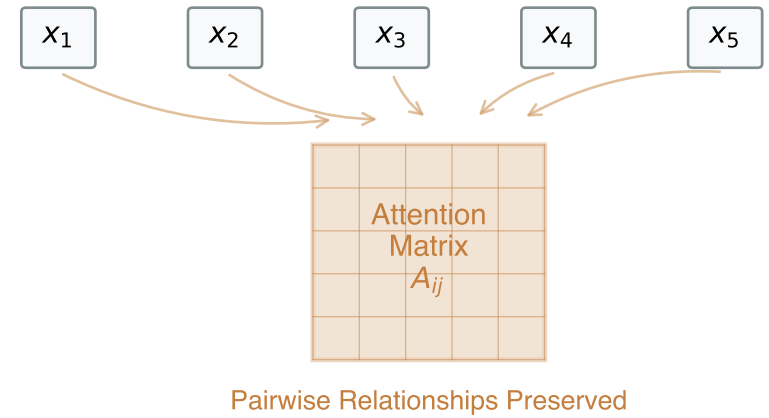


## A. Sequential Compression (RNN / LSTM)



$$h_t = f(h_{t-1}, x_t) \text{ — Sequential update}$$

## B. Lossless Covariance (Transformer)



$$A = \text{softmax}\left(\frac{QK^T}{\sqrt{d}}\right) \text{ — All-pairs attention}$$

## C. Biological Mechanism Integration Across Architectures

